

A Review on Wearable Technologies in Early Detection of Lymphedema

Shahariar H. Ronok and Md. Robiul Islam

Abstract—Lymphedema is a condition that causes swelling in the arms or legs due to the accumulation of lymphatic fluid, and early detection is crucial to prevent the condition from worsening. Wearable technologies in the early detection of lymphedema have been gaining attention in recent years. In several studies, wearable technologies, including sensors as well as machine learning algorithms, have been investigated for their viability and efficacy in the early identification of lymphedema. Yet, public acceptance and adoption of these devices remain a challenge. Moreover, factors such as cost, comfort, and user-friendliness need to be addressed to increase adoption rates. This study found wearable technologies research on lymphedema early detection shows promising results and has the potential to revolutionize patient care. However, continued efforts to improve device design and increase public awareness and education can help overcome the barriers to adoption and facilitate the widespread use of wearable technology for lymphedema early detection.

Index Terms—Lymphedema, wearable technologies, remote monitoring, early detection.

I. INTRODUCTION

THE excessive accumulation of lymph in the limbs is a chronic condition that causes swelling in a number of bodily parts which causes lymphedema disease. This swelling results from the buildup of lymph fluid in the organs and is frequently brought on by harm to the lymph node system. Lymphedema comes in two varieties: primary and secondary [1], [2]. Abnormalities in the lymphatic system bring on an uncommon genetic disease called primary lymphedema, which may grow during adolescence or adulthood or be present at birth. The more prevalent type of the condition, in contrast, is known as secondary lymphedema, and it is usually brought on by harm to the lymph glands as a result of treatment with radiation, surgical infection, injury, or additional health issues [1], [3]. Combinations of treatments intended to reduce swelling, enhance lymphatic flow, and guard against infection are frequently used in the treatment of lymphedema. Compression clothing, manual drainage of lymph therapy, exercise, skin maintenance, and surgery are all potential treatments.

While managing the symptoms and improving quality of life while maintaining lymphedema can be difficult, it is doable with the right management and treatment. If you have undergone cancer treatment or have other risk factors, it is critical to be conscious of the potential hazards of lymphedema

and to notify your healthcare provider of any symptoms. It is possible to avoid or reduce the adverse effects of lymphedema with early detection and therapy.

An inventive method of remotely monitoring patients' health status is through wearable technology. It entails taking advantage of wearable devices that can gather and send information like blood pressure, heart rate, and levels of activity to healthcare experts for analysis and eventual intervention. Access to treatment is made easier thanks to distant monitoring using wearable technology. Patients who reside in rural regions or lack transportation to a healthcare center can receive prompt interventions from medical staff members wherever they are. The frequency of expensive emergency room trips and hospital readmissions may also be decreased [4]. Continuous observation enables early identification of health problems and quicker intervention, providing patients who have chronic conditions crucial real-time feedback on their health status [5].

II. LYMPHEDEMA

Lymphedema is a chronic, disabling disease that develops when the lymphatic system, which is in charge of removing extra fluid from tissues and redistributing it throughout the bloodstream, is harmed or blocked. This leads to a buildup of lymph fluid in the affected area, typically in the limbs, causing swelling, pain, and reduced mobility. Primary lymphedema is a rare genetic condition that affects the development of the lymphatic system, while secondary lymphedema is typically the result of damage to the lymphatic system due to surgery, radiation therapy, infection, or trauma.

Early detection is essential in the treatment of lymphedema, as it allows for prompt intervention and better outcomes. Patients who receive early treatment for lymphedema are more likely to experience a reduction in swelling, pain, and other symptoms. Treatment typically involves a combination of therapies, such as compression garments, exercise, and manual lymphatic drainage, which can help reduce swelling and improve lymphatic function [1], [3], [6]. Early intervention can also prevent the need for more invasive and expensive treatments down the line.

III. WEARABLE TECHNOLOGIES IN HEALTHCARE

Electronic gadgets that can be worn on the body, such as watches, glasses, fitness monitors, and other accessories, are known as wearable technologies. One of the main advantages of wearable technologies is their portability [7]. Another benefit is their ability to provide real-time feedback. Unlike traditional devices that require users to manually input data, wearable devices can automatically collect and transmit data to other devices [4], [5].

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IV. WEARABLE TECHNOLOGIES IN DETECTION OF LYMPHEDEMA

Innovative methods to track the disease's development include wearable interventions. Electronic gadgets that can be sewn into clothing or attached to the flesh are known as wearable technologies [8]. Affordable, simple to use, and space-saving, wearable treatments are becoming more popular, though research on this novel approach is still emerging [9]–[14]. A comprehensive summary of clinical and technological investigations is cataloged in Table I and Table II.

A. Yanmin *et al.*

Yanmin *et al.* [9] employed low-ductility massage belts along with an elastic sensor module and a signal processing module to both monitor and provide compression treatment for lymphedema. The forearm low-ductility massage belts are positioned inside the wrist in order of decreasing ductility. With the help of a motor or compressed air, each massage belt can mimic manual lymphatic drainage by applying a set output stress load to each portion of the upper limb (Fig. 1(a)).

B. Fallahzadeh *et al.*

The wearable platform suggested by Fallahzadeh *et al.* [10] is low-power, context-aware, and real-time for continuous remote edema monitoring (Fig. 1(b)–(c)). The system evaluates foot edema by monitoring changes in the foot radius of curvature and the present body posture. The sensor demonstrated an R^2 of 0.87 for the regression model, an intraclass correlation of 0.97, and over 96% accuracy in activity monitoring across 15 subjects.

C. Bethencourt *et al.*

Bethencourt *et al.* [11] examined the creation of a clinical trial using a connected medical device intended to detect lymphedema after axillary lymph node dissection (ALND) (Fig. 1(d)). Stochastic transformation and sensitivity studies were used to develop measurement protocols on a digital twin simulator.

Using the growth function f , a basic model was developed to predict the theoretical circumference of the arm at position i and time t :

$$\begin{aligned} y_t^i &= M_1(t, y_0^i, f, t_0, r^i, y_\infty^i, \epsilon_t^i) \\ &= \begin{cases} y_0^i + \epsilon_t^i, & t < t_0 \\ f(t, y_0^i, r^i, y_\infty^i) + \epsilon_t^i, & t \geq t_0 \end{cases} \end{aligned} \quad (1)$$

D. Huang *et al.*

Huang *et al.* [12] demonstrated wireless epidermal sensors attached to an expandable balloon to assess simulated lymphedema (Fig. 2(a)). Extending sensor 1 in the x -direction by 29.1% causes f_0 to drop by 22.7 MHz, with sensor 2 shifting resonance by 22.5 MHz. The frequency response is governed by:

$$f = \frac{1}{2\pi\sqrt{LC}} \quad (2)$$

E. Kato *et al.*

Kato *et al.* [13] created a prototype to quantify fluid retention in a lymphedema region using an FSR sensor embedded inside an air pack (Fig. 2(b)). By altering air pressure in five stages (0–40 mmHg), a tight correlation ($r = 0.998$) was established between reaction force and moisture content.

F. Zhang *et al.*

Zhang *et al.* [14] utilized agar-water tissue phantoms to validate a wearable water content sensor combining ultrasound and magnetic sensing (Fig. 2(c)). Ultrasound velocity v is determined by:

$$v = \frac{d}{\text{TOF}} \quad (3)$$

V. DISCUSSION

The studies that have investigated wearable technologies in monitoring lymphedema are still in their exploratory stages, typically characterized by small cohort sizes. These studies frequently lack longitudinal evaluation of real-time monitoring impact on upper arm lymphedema (UAL) outcomes. Furthermore, few interventions have undergone full clinical certification or commercial translation.

VI. FUTURE ASPECTS & INNOVATION

Wearable solutions such as the Lymphometer illustrate the growing demand for home-based limb circumference surveillance. Moving forward, combining continuous strain transducers with computer vision and machine learning models will allow clinicians to track fluid shifts alongside limb motion dynamics. Educating patients on proper adherence and integrating continuous telemetry into electronic health records will be crucial for clinical adoption.

VII. LIMITATIONS

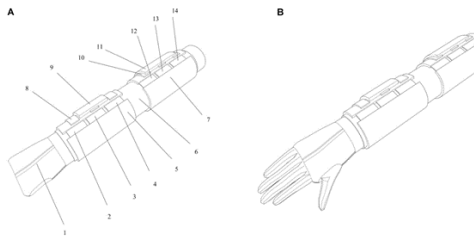
This literature review focuses primarily on volumetric and fluid variations induced by lymphatic accumulation. Studies published between January 1, 2005, and September 30, 2022, in the English language were evaluated. Older legacy techniques were omitted to focus on modern sensing paradigms.

VIII. CONCLUSION

Wearable sensors and portable 3D imaging offer promising avenues for decentralized, client-centered lymphedema monitoring. Advancements in signal processing and flexible sensor integration will enable earlier detection windows, minimizing complications and improving long-term quality of life for cancer survivors.

TABLE I
SUMMARY OF STUDY CHARACTERISTICS: WEARABLE SENSORS & CLINICAL SYSTEMS (PART I)

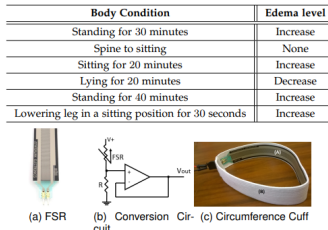
Author (Date), Loc.	Aim of the Study	Method	Indicators	Technology	Participants (n; Condition)	Outcome / Result
White et al. (2021), USA [15]	Determine feasibility of portable scanning in UAL screening & demographic association.	3D scanning	Optical + IR + Motion	iPad Mini 2 (Apple)	$n = 21$; Female; BCRL patients	Chemo patients and relative volume change ($< 5\%$) were correlated ($p = 0.0272$).
Yanmin et al. (2021), China [9]	Optimize portable wearable device for remote monitoring & compression therapy.	Circumferential elasticity change	Elasticity	Elastic sensor module	$n = 1$; Female; 20–30 yrs; Grade 0 lymphedema	More accurate than manual tape measurement; ensures consistent placement. No clinical validation.
Vitali et al. (2021), Italy [16]	Create virtual framework for unbiased clinician assessment during treatment.	3D scanning	Optical + IR + Motion	Structure Sensor + iPad	$n = 8$; M: 6, F: 2; Age 24 yrs; Healthy	Average difference of 2 mm; errors found negligible by clinical evaluators.
Donahue et al. (2020), USA [17]	Evaluate whether BCRL abnormalities affect both hemispheres beyond standard areas.	Bioimpedance spectroscopy	Low Frequency Current	ImpediMed L-Dex@U400	$n = 66$; F; BCRL: 33 (54.1 yrs); Healthy: 33 (49.4 yrs)	Relaxation times elevated for superficial tissue and deep muscle ($p = 0.04$).
Binkley et al. (2020), USA [18]	Validate clinical usage of LymphaTech in screening upper arm lymphedema.	3D scanning	Optical + IR + Motion	iPad (Apple)	$n = 66$; Female; Mean age 55 yrs; BCSPs	ICC ≤ 0.99 with Perometer. Measurement bias was between 38.0–40.7 ml.
Lu et al. (2019), USA [19]	Demonstrate Kinect IR system capability to assess UAL cost-effectively.	Single-point 3D scanning	Optical + IR + Motion	Kinect IR sensor	$n = 73$; Female; Age 18+ yrs; BCSPs	Moderate agreement ($\kappa = 0.5475$) at 200 ml difference; clinically valid tool for severe cases.
Kato et al. (2019), Japan [13]	Develop wearable device estimating fluid content using pressure.	Pressure sensing	Pressure	FSR sensor	$n = 11$; Female; 20–30 yrs; Healthy	Regression coefficient difference was significant ($P < 0.003$); Moisture D correlation $r = 0.998$.
Hashemi et al. (2018), Canada [20]	Ensure reliability and cost-effectiveness of tissue assessment.	Ultrasound elastography	Ultrasound	CIRS elastography phantom	$n = 7$; Female; 54–81 yrs; Stage 2 lymphedema	Significant Wilcoxon test (p for skin = 1.24×10^{-5} , muscle = 8.11×10^{-7}).
Zhang et al. (2018), USA [14]	Wearable sensor device for convenient home-based fluid accumulation tracking.	Ultrasound velocity shift	Ultrasound velocity	Wearable water sensor	$n = 1$; Male; Not mentioned	Ultrasound velocity lower (1521–1629 m/s) in upper leg region.
Buffa et al. (2015), Italy [21]	Validate 3D acquisition system with SkanLab software to estimate arm volume.	360° 3D imaging	Optical + IR + Motion	Kinect sensor + frame	$n = 30$; F: 15, M: 15; Age 19–60 yrs	Arm volume bias -0.6% (-9.9 ml); limits of agreement -2.6% to 1.4% ; inter-rater > 0.99 .
Huang et al. (2014), USA [12]	Flexible device laminated to skin for wireless surface strain & dielectric sensing.	RF change (1 MHz–1 GHz)	Moisture + Strain	Epidermal sensor + analyzer	Balloon phantom; N/A	Precision = 1.1 for moisture measurement; strain detected up to 1.3%.
Zhou et al. (2019), USA [22]	Validate 3DIS in detecting filarial lymphedema in field conditions.	3D imaging	Optical + IR + Motion	Structure Sensor + iPad	$n = 41$; F: 33, M: 8; Age 56; Lower lymphedema	Successful field screening; volume variation 1.7% (left leg) and 2.2% (right leg).
Williams et al. (2018), USA [23]	Device to differentiate between lymphedema stages using pitting edema assessment.	Mechanical properties	RGB Image	Camera + air flow	Tissue phantoms; N/A	Area-vs-time data showed distinct differences across all four lymphedema stages.



(a) Yanmin et al. [9]



(b) Ankle monitor [10]



(c) FSR cuff [10]



(d) Lymphometer [11]

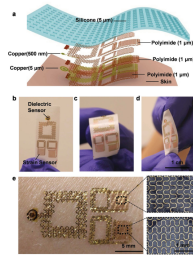
Fig. 1. Wearable physical sensor and sleeve prototypes across reviewed studies: (a) pneumatic massage sleeve, (b) ankle motion tracker, (c) FSR circumference cuff, and (d) connected sleeve prototype.

REFERENCES

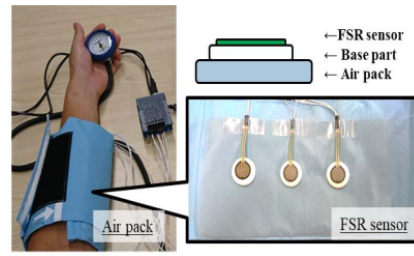
- [1] K. Rincon, P. Shah, J. Ramella-Roman, and S. Bhansali, "A review of engineering approaches for lymphedema detection," *IEEE reviews in biomedical engineering*, vol. 9, pp. 79–90, 2016.
- [2] A. K. Greene, "Epidemiology and morbidity of lymphedema," *Lymphedema: Presentation, Diagnosis, and Treatment*, pp. 33–44, 2015.

TABLE II
SUMMARY OF STUDY CHARACTERISTICS: 3D SCANNING, BIOIMPEDANCE, AND DIELECTRIC METHODS (PART II)

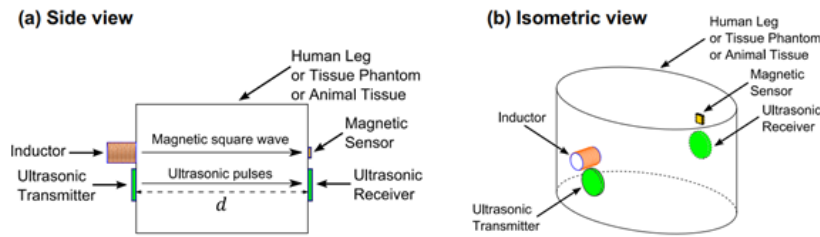
Author (Date), Loc.	Aim of the Study	Method	Indicators	Technology	Participants (n; Condition)	Outcome / Result
Fallahzadeh et al. (2016), USA [10]	Real-time low-powered wearable device to track edema remotely.	Stretch sensor elongation	Electrical Resistance	Cuff + longitudinal FSR	$n = 10$; Age 22–31 yrs; Healthy	$R^2 = 0.87$ and $ICC = 0.971$; > 96% measurement accuracy achieved.
Bethencourt et al. (2021), France [11]	Client-centric lymphedema monitoring system with patient-to-doctor protocol.	Stretch sensor elongation	Electrical Resistance	Stretch sensor in sleeve	$n = 1$; Female; BCRL patient	Implemented digital twin protocol for home surveillance; accuracy metrics omitted.
Karakashian et al. (2018), UK [24]	Evaluate 3D depth camera suitability for clinical screening of lymphedematous arms.	360° 3D imaging	Optical + IR + Motion	ASUS Xtion Pro 3D	$n = 24$; Female; 29–76 yrs; Unilateral lymphedema	Significant difference ($p < 0.05$) between affected and healthy arms; $ICC = 0.957$.
Erdogan et al. (2019), Turkey [25]	Diagnose and stage lymphedema using sonoelastography.	Elastography	Ultrasound	Acuson S 3000 US®	$n = 36$; Female; Age 50.8 yrs; Stages 1 and 2	Circumference and elastography correlated ($r = 0.471$, $p = 0.004$); correlated with L-Dex.
Cau et al. (2018), Italy [26]	Validate laser scanning system in screening upper arm lymphedema.	3D scanning	Laser	Rodin4D scanner	$n = 200$; Female; 64.7 yrs; UAL patients	Correlation ($r = 0.738$, $P < 0.05$) with manual measurement with bias -0.09 dm^3 .
Ohberg et al. (2014), Sweden [27]	Analyze 3D acquisition method compared with traditional techniques.	3D scanning	Optical + IR + Motion	3 Kinect v1 sensors	$n = 25$; Female; 60.5 yrs; Lymphedema	Overestimates volume by 45.25 ml ($p = 0.270$) compared to water displacement.
Hoevenaren et al. (2016), Netherlands [28]	Verify 3D volumetric scanning as clinical tool in upper arm assessment.	Stereophotogrammetry	RGB Image	15 cameras (5 pods)	$n = 18$; Female; 56.5 yrs; Hand edema	Hand edema and control had significantly different volumes ($\chi^2 = 18.9$, $p < 0.001$).
Verhulst et al. (2017), Netherlands [29]	Validate reliability of 3D stereophotogrammetry for upper arm volume measurement.	Stereophotogrammetry	RGB Image	15 cameras (5 pods)	$n = 10$; M: 5, F: 5; Age 28.6 yrs; Healthy	Intrarater reliability was 0.96 (hand) and 0.99 (forearm); interrater was 0.98 and 0.99.
Koelmeyer et al. (2021), Australia [30]	Test feasibility of Bioimpedance System (BIS) home monitoring for high-risk patients.	Bioimpedance spectroscopy	Low Frequency Current	SOZO® (ImpediMed)	$n = 20$; Female; 18–85 yrs; Post-BCRL	Adherence was 74%; successfully identified 5 patients with lymphedema (L-Dex > +6.5).
Bakar et al. (2018), Turkey [31]	Ascertain specificity and sensitivity of local tissue water (LTW) in staging.	Tissue dielectric constant	300 MHz frequency	Moisture Meter-D	$n = 63$; Female; 53.3 yrs (latent), 54.5 yrs	Absolute LTW and ratio significant ($p < 0.001$). Sensitivity = 65%, specificity = 94%.
Lee et al. (2011), Australia [32]	Determine Perometer validity for clinical assessment of lymphedema.	3D Perometry	Infrared (IR)	Perometer	$n = 40$; Female; 56 yrs; Lymphedema: 20	ICC was 0.989 (95% CI); inter-rater reliability was 0.993 with 7.5% bias.
Hameeteman et al. (2016), Netherlands [33]	Evaluate 3D stereophotogrammetry reliability in upper arm assessment.	Stereophotogrammetry	RGB Image	15 cameras (5 pods)	$n = 11$; Female; Age 54 yrs; Post-BCRL	Correlation with water displacement $r = 0.99$ ($p < 0.01$).



(a) Conformal epidermal sensor [12]



(b) Pressure cuff with air pack [13]



(c) Ultrasonic & magnetic sensor [14]

Fig. 2. Transducer modalities: (a) Huang et al. conformal epidermal sensor, (b) Kato et al. reaction force pressure pack, and (c) Zhang et al. acoustic water content measurement transducer.

- [3] S. G. Rockson, "Lymphedema," *Vascular Medicine*, vol. 21, no. 1, pp. 77–81, 2016.
- [4] J. Dunn, R. Runge, and M. Snyder, "Wearables and the medical revolution," *Personalized medicine*, vol. 15, no. 5, pp. 429–448, 2018.
- [5] M. Wu and J. Luo, "Wearable technology applications in healthcare: a literature review," *Online J. Nurs. Inform.*, vol. 23, no. 3, 2019.
- [6] P. Kamali and S. J. Lin, "Lymphedema: presentation, diagnosis, and treatment," *Plastic and reconstructive surgery*, vol. 137, no. 5, pp. 1654–1655, 2016.
- [7] A. Rajkumar, F. Vulpi, S. R. Bethi, H. K. Wazir, P. Raghavan, and V. Kapila, "Wearable inertial sensors for range of motion assessment," *IEEE sensors journal*, vol. 20, no. 7, pp. 3777–3787, 2019.

- [8] A. Hayes, "The ins and outs of wearable technology," Jul 2022. [Online]. Available: <https://www.investopedia.com/terms/w/wearable-technology.asp>
- [9] X. Yanmin, Z. Xuyang, Y. Wen, Y. Suihuai, and L. Sinan, "Study on the design and optimization of a portable monitoring and auxiliary treatment device for upper extremity lymphedema—focus on the rehabilitation function of the device," *Frontiers in Bioengineering and Biotechnology*, vol. 9, p. 656716, 2021.
- [10] R. Fallahzadeh, Y. Ma, and H. Ghasemzadeh, "Context-aware system design for remote health monitoring: An application to continuous edema assessment," *IEEE Transactions on Mobile Computing*, vol. 16, no. 8, pp. 2159–2173, 2016.
- [11] L. Bethencourt, W. Dabachine, V. Dejoui, Z. Lalmiche, K. Neuberger, I. Ibnouhsein, S. Chereau, C. Mathelin, N. Savy, P. Saint Pierre *et al.*, "Guiding measurement protocols of connected medical devices using digital twins: A statistical methodology applied to detecting and monitoring lymphedema," *IEEE Access*, vol. 9, pp. 39 444–39 465, 2021.
- [12] X. Huang, Y. Liu, H. Cheng, W.-J. Shin, J. A. Fan, Z. Liu, C.-J. Lu, G.-W. Kong, K. Chen, D. Patnaik *et al.*, "Materials and designs for wireless epidermal sensors of hydration and strain," *Advanced Functional Materials*, vol. 24, no. 25, pp. 3846–3854, 2014.
- [13] D. Kato, R. Tajika, K. Mototani, M. Makikawa, M. Tsukuda, A. Ishii, and S. Okada, "Estimation of subcutaneous tissue state of lymphedema using pressure sensor," in *2019 IEEE 1st Global Conference on Life Sciences and Technologies (LifeTech)*. IEEE, 2019, pp. 76–77.
- [14] S. Zhang, R. Rajamani, and A. S. Sezen, "Wearable water content sensor based on ultrasound and magnetic sensing," *Annals of Biomedical Engineering*, vol. 46, pp. 2079–2090, 2018.
- [15] J. White *et al.*, "The use of a portable three-dimensional scanning device to measure limb volume in breast cancer-related lymphedema," *Lymphatic Research and Biology*, vol. 19, no. 4, pp. 365–372, 2021.
- [16] A. Vitali, G. Togni, D. Regazzoni, C. Rizzi, and G. Molinero, "A virtual environment to evaluate the arm volume for lymphedema affected patients," *Computer Methods and Programs in Biomedicine*, vol. 198, p. 105795, 2021.
- [17] P. Donahue, R. Crescenzi, C. Lee, M. Garza, N. J. Patel, K. J. Petersen, and M. J. Donahue, "Magnetic resonance imaging and bioimpedance evaluation of lymphatic abnormalities in patients with breast cancer treatment-related lymphedema," *Breast cancer research and treatment*, vol. 183, no. 1, pp. 83–94, 2020.
- [18] J. M. Binkley, M. J. Weiler, N. Frank, L. Bober, J. B. Dixon, and P. W. Stratford, "Assessing arm volume in people during and after treatment for breast cancer: reliability and convergent validity of the lymphatech system," *Physical Therapy*, vol. 100, no. 3, pp. 457–467, 2020.
- [19] I. M. Lu and J. B. Dixon, "Assessment of upper extremity swelling among breast cancer survivors with a commercial infrared sensor," *Lymphatic Research and Biology*, vol. 17, no. 4, pp. 424–433, 2019.
- [20] H. S. Hashemi, S. Fallone, M. Boily, A. Towers, R. D. Kilgour, and H. Rivaz, "Assessment of mechanical properties of tissue in breast cancer-related lymphedema using ultrasound elastography," *IEEE transactions on ultrasonics, ferroelectrics, and frequency control*, vol. 66, no. 3, pp. 541–550, 2018.
- [21] R. Buffa, E. Mereu, P. Lussu, V. Succa, T. Pisanu, F. Buffa, and E. Marini, "A new, effective and low-cost three-dimensional approach for the estimation of upper-limb volume," *sensors*, vol. 15, no. 6, pp. 12 342–12 357, 2015.
- [22] C. Zhou, C. Yahathugoda, L. De Silva, U. Rathnapala, G. Owen, M. Weerasooriya, R. U. Rao, G. J. Weil, and P. J. Budge, "Portable infrared imaging for longitudinal limb volume monitoring in patients with lymphatic filariasis," *PLoS neglected tropical diseases*, vol. 13, no. 10, p. e0007762, 2019.
- [23] K. Williams, M. Han, S. Hardin, S. George, and J. Yao, "Aero: An objective peripheral edema measurement device," in *2018 40th Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC)*. IEEE, 2018, pp. 5914–5917.
- [24] K. Karakashian, L. Shaban, C. Pike, and R. van Loon, "Investigation of shape with patients suffering from unilateral lymphoedema," *Annals of Biomedical Engineering*, vol. 46, no. 1, pp. 108–121, 2018.
- [25] Z. Erdogan Iyigun, F. Agacayak, A. S. Ilgun, F. Elbuken Celebi, C. Ordu, G. Alco, A. Ozturk, T. Duymaz, F. Aktepe, and V. Ozmen, "The role of elastography in diagnosis and staging of breast cancer-related lymphedema," *Lymphatic research and biology*, vol. 17, no. 3, pp. 334–339, 2019.
- [26] N. Cau, M. Galli, V. Cimolin, A. Grossi, I. Battarin, G. Puleo, A. Balzarini, and A. Caraceni, "Quantitative comparison between the laser scanner three-dimensional method and the circumferential method for evaluation of arm volume in patients with lymphedema," *Journal of Vascular Surgery: Venous and Lymphatic Disorders*, vol. 6, no. 1, pp. 96–103, 2018.
- [27] F. Öhberg, A. Zachrisson, and Å. Holmner-Rocklöv, "Three-dimensional camera system for measuring arm volume in women with lymphedema following breast cancer treatment," *Lymphatic research and biology*, vol. 12, no. 4, pp. 267–274, 2014.
- [28] I. A. Hoevenaren, A. Verhulst, M. Hameeteman, R. Vreeken, T. J. Maal, and D. Ulrich, "Three-dimensional stereophotogrammetry as an accurate tool for analyzing lymphedema of the hand," *JPRAS Open*, vol. 10, pp. 40–46, 2016.
- [29] A. Verhulst, T. Wesselijs, H. Glas, R. Vreeken, D. Ulrich, and T. Maal, "Accuracy and reproducibility of a newly developed tool for volume measurements of the arm using 3d stereophotogrammetry," *Journal of Plastic, Reconstructive & Aesthetic Surgery*, vol. 70, no. 12, pp. 1753–1759, 2017.
- [30] L. A. Koelmeyer, E. Moloney, J. Boyages, K. A. Sherman, and C. M. Dean, "Prospective surveillance model in the home for breast cancer-related lymphoedema: A feasibility study," *Breast Cancer Research and Treatment*, vol. 185, no. 2, pp. 401–412, 2021.
- [31] Y. Bakar, A. Tuğral, and Ü. Üyetürk, "Measurement of local tissue water in patients with breast cancer-related lymphedema," *Lymphatic Research and Biology*, vol. 16, no. 2, pp. 160–164, 2018.
- [32] M. J. Lee, R. A. Boland, S. Czerniec, and S. L. Kilbreath, "Reliability and concurrent validity of the perometer for measuring hand volume in women with and without lymphedema," *Lymphatic research and Biology*, vol. 9, no. 1, pp. 13–18, 2011.
- [33] M. Hameeteman, A. C. Verhulst, R. D. Vreeken, T. J. Maal, and D. J. Ulrich, "3d stereophotogrammetry in upper-extremity lymphedema: an accurate diagnostic method," *Journal of Plastic, Reconstructive & Aesthetic Surgery*, vol. 69, no. 2, pp. 241–247, 2016.